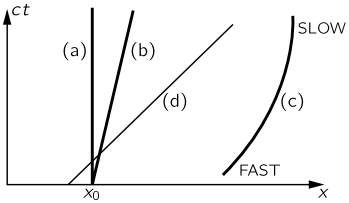

Figure 5-1 Three particle paths in space-time: (a) a particle at rest at $x = x_0$; (b) a particle which starts as $x = x_0$ and moves with constant speed; (c) a particle which starts at high speed but slows down.

What about light? Light travels at the speed c, and that would be represented by a line having a certain fixed slope (Figure 5-1[d]).

Now according to our new idea, if a given event occurs to a particle, say if it suddenly disintegrates at a certain space-time point into two new ones which follow some new tracks, and this interesting event occurred at a certain value of x and a certain value of t, then we would expect that, if this makes any sense, we just have to take a new pair of axes and turn them, and that will give us the new t and the new x in our new system, as shown in Figure 5-2(a). But this is wrong, because Eq. (5.1) is not *exactly* the same mathematical transformation as Eq. (5.2). Note, for example, the difference in sign between the two, and the fact that one is written in terms of cos θ and sin θ, while the other is written with algebraic quantities. (Of course, it is not impossible that the algebraic quantities could be written as cosine and sine, but actually they cannot.) But still, the two expressions *are* very similar.

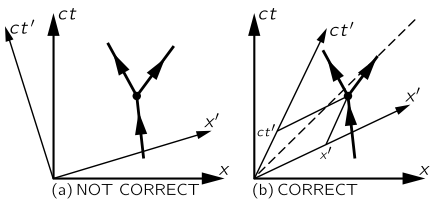

Figure 5-2 Two views of a disintegrating particle.